BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The lymphatic system

The written notes for the lymphatic system lecture: the one-way lymph pathway, the structure of lymphatic vessels and lymph nodes, the primary and secondary lymphoid organs, and the common lymphatic disorders. Module 4 competencies 105 to 108, with eight worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 105. Lymph pathway

In your course notes: Module 4 packet, Chapter 2, competency 105 (page 18).

Dr. Rennie's video for this part: Lymphatic system concept video.

You should be able to: Define lymph and trace its one-way route from lymphatic capillaries through collecting vessels, nodes, trunks, and ducts back to the subclavian veins.

Blood capillaries leak. All day long, fluid filters out of them into the spaces between cells, and that fluid is called interstitial fluid. The blood capillaries take most of it back, but some fluid and a small amount of plasma protein stay behind in the tissues. The lymphatic system collects this leftover fluid, about 3 liters a day, and returns it to the blood. That keeps fluid from building up in the tissues and keeps blood volume steady.

Once interstitial fluid enters a lymphatic vessel, it is called lymph. Lymph is clear and watery. It carries water, plasma proteins, and whatever else it picks up in the tissues, such as cell debris, bacteria, and sometimes cancer cells. Lymph coming from the small intestine after a fatty meal looks milky white. This fat-rich lymph is called chyle.

The fluids and the system

Lymphatic system
: A network of vessels, tissues, and organs that drains tissue fluid and houses the body's defenses.

Interstitial fluid
: The fluid in the spaces between cells, filtered out of the blood capillaries. In the tissue spaces it is still interstitial fluid, not lymph.

Lymph
: Interstitial fluid once it has entered a lymphatic vessel. It becomes lymph the moment it enters a lymphatic capillary. Clear and watery.

Chyle
: Milky, fat-rich lymph coming from the lacteals of the small intestine.

Three functions
: Returns excess interstitial fluid to the blood, absorbs dietary fats from the small intestine, and houses the immune defenses.

The one-way route, in order

The lymphatic system is a one-way system. It does not form a loop the way blood vessels do. Lymph starts in the tissues and flows only toward the heart. As it travels, small vessels join into larger and larger vessels, the way small streams join to form a river. On the way, the lymph

passes through lymph nodes, which filter it before it reaches the blood.

1. Lymphatic capillaries: blind-ended tubes in the tissues take up excess interstitial fluid. The fluid is now lymph.
2. Lymphatic collecting vessels: larger vessels with valves carry the lymph onward and keep it moving one way.
3. Lymph nodes: nodes spaced along the collecting vessels filter the lymph. Lymph usually passes through several nodes in a row.
4. Lymphatic trunks: collecting vessels merge into large trunks. Each trunk drains one major region of the body.
5. Lymphatic ducts: the trunks empty into one of two ducts, the right lymphatic duct or the larger thoracic duct.
6. Subclavian veins: each duct empties into the blood where the subclavian vein meets the internal jugular vein on its own side. At this point lymph becomes part of the blood plasma again.

- Worked example from the note sheet, starting at the left foot: a lymphatic capillary in the foot, a collecting vessel with valves running up the leg, an inguinal node in the groin, the left lumbar trunk running up the back wall of the abdomen, the cisterna chyli, the thoracic duct up through the thorax, and finally the junction of the left subclavian and left internal jugular veins. Every arrow points toward the heart and none point back.

The nine trunks

There are nine lymphatic trunks. Each one is named for the region it drains. Four of them come in left and right pairs, and one, the intestinal trunk, is single.

Trunk	Region drained	Empties into
Lumbar (left and right)	Lower limbs, pelvis, and abdominal wall	Thoracic duct, at the cisterna chyli
Intestinal (single)	Digestive organs of the abdomen	Thoracic duct, at the cisterna chyli
Bronchomediastinal (left and right)	Thoracic organs and the thoracic wall	Left: thoracic duct. Right: right lymphatic duct
Subclavian (left and right)	Upper limbs	Left: thoracic duct. Right: right lymphatic duct
Jugular (left and right)	Head and neck	Left: thoracic duct. Right: right lymphatic duct

The two ducts

The right lymphatic duct is short and small. It drains the right upper limb and the right side of the head, neck, and thorax. The thoracic duct is the largest lymphatic vessel in the body. It drains everything else, which is about three quarters of the body. The thoracic duct begins in the abdomen as a sac called the cisterna chyli, which sits in front of the upper lumbar vertebrae. From there it passes up through the diaphragm with the aorta and climbs through the thorax beside the vertebral column.

Both ducts end in the base of the neck. Each one empties into the junction of the subclavian vein and the internal jugular vein on its own side. The right lymphatic duct empties on the right side. The thoracic duct empties on the left side.

Feature	Right lymphatic duct	Thoracic duct
Size	Short and small	Largest lymphatic vessel
Drains	Right upper limb and the right side of the head, neck, and thorax	Everything else, about three quarters of the body
Trunks it receives	Right jugular, right subclavian, right bronchomediastinal	Both lumbar, intestinal, left jugular, left subclavian, left bronchomediastinal
Starts at	The right side of the base of the neck	The cisterna chyli, a sac in the abdomen
Empties into	Junction of the right subclavian and right internal jugular veins	Junction of the left subclavian and left internal jugular veins

- Memory tip: split the body into four quarters. The right upper quarter (right arm, right side of the head, neck, and chest) drains into the right lymphatic duct. The other three quarters drain into the thoracic duct. The right leg is thoracic duct territory, not right lymphatic duct territory.

How lymph moves without a pump

The lymphatic system has no heart to push the lymph. Lymph is moved by forces from the structures around the vessels, and the valves keep it from sliding backward.

- Skeletal muscle pump: contracting muscles squeeze the vessels that run between them.
- Respiratory pump: pressure changes in the thorax with each breath pull lymph upward.
- Smooth muscle in the walls of the larger lymphatic vessels contracts in rhythm.
- Pulsing of nearby arteries presses on the vessels.
- Valves in the collecting vessels keep all of this movement going one way, toward the heart.

Competency 106. Vessel and node structure

In your course notes: Module 4 packet, Chapter 2, competency 106 (page 20).

Dr. Rennie's video for this part: Lymphatic system concept video.

You should be able to: Describe the structure of lymphatic capillaries, valved collecting vessels, and a lymph node including capsule, cortex, medulla, hilum, and afferent and efferent vessels.

Lymphatic capillaries

Lymphatic capillaries are the smallest lymphatic vessels. They lie in the tissues, woven between the blood capillaries. Each one is a thin tube that is closed at one end, so it is called blind-ended. Its wall is a single layer of endothelial cells.

The endothelial cells of a lymphatic capillary overlap at their edges. These loose, overlapping endothelial cells act as one-way flap valves. When fluid builds up in the tissue, the flaps push open and fluid flows in. When pressure inside the capillary rises, the flaps press shut, so the fluid cannot flow back out. Fine collagen strands called anchoring filaments tie the endothelial cells to the surrounding tissue. When the tissue swells, the filaments pull on the cells and hold the capillary open instead of letting it collapse.

Because the gaps between the cells are large, lymphatic capillaries are more permeable than blood capillaries. Proteins, cell debris, bacteria, and even cancer cells can get in. Lymphatic capillaries are found in nearly every tissue. They are absent from the central nervous system (the brain and spinal cord), bone and red bone marrow, and avascular tissues such as cartilage, the epidermis, and the cornea.

Feature	Lymphatic capillary	Blood capillary
Ends	Blind (closed) at one end	Open at both ends, part of a loop
Size	Wider	Narrower
Wall	Overlapping endothelial cells act as one-way flap valves	Endothelial cells joined more tightly
Held open by	Anchoring filaments tied to the surrounding tissue	Pressure of the blood inside
What gets in	Fluid, proteins, cells, pathogens, and even cancer cells	Mostly small molecules
Flow	One way, away from the tissues	Through, from the artery side to the vein side

- In the small intestine, each villus has a special lymphatic capillary in its center called a lacteal. Lacteals absorb dietary fats, which are packaged in particles too large to enter blood capillaries. The fat turns the lymph milky, and this lymph (chyle) drains to the cisterna chyli and up the thoracic duct.

Collecting vessels and their valves

Lymphatic capillaries join to form lymphatic collecting vessels. These are built like small veins. Their walls have the same three layers as a vein, but the layers are thinner. Collecting vessels have more internal valves than veins do. These lymphatic valves are one-way

flaps, so lymph can move forward but cannot flow backward. When a collecting vessel is full, the stretches between valves bulge, which gives the vessel a beaded look.

- Superficial collecting vessels run in the subcutaneous layer of the skin alongside the superficial veins. Deep collecting vessels run alongside the deep arteries.
- Along their length, the collecting vessels pass through lymph nodes.

The lymph node

A lymph node is a small, bean-shaped organ that filters lymph. Hundreds of nodes lie along the collecting vessels. They are clustered in the neck (cervical nodes), the armpit (axillary nodes), and the groin (inguinal nodes), and there are many more deep in the thorax and abdomen.

The outside of a node is a capsule of dense connective tissue. Strands of connective tissue called trabeculae extend inward from the capsule and divide the node into compartments. Inside, a mesh of reticular fibers holds the lymphocytes and macrophages in place.

The node has two regions. The outer region is the cortex. The outer cortex holds rounded clusters of lymphocytes called lymphatic follicles, which are mostly B cells. Many follicles have a pale germinal center where B cells multiply. The deep cortex is mostly T cells. The inner region is the medulla. It contains medullary cords, strands of lymphocytes and plasma cells, separated by medullary sinuses, open channels lined with macrophages.

One side of the node is indented. This is the hilum. Blood vessels enter and leave the node at the hilum, and lymph leaves there too. Afferent lymphatic vessels carry lymph into the node. There are several of them, and they enter at many points on the rounded (convex) surface through the capsule. Efferent lymphatic vessels carry filtered lymph out of the node, and they leave only at the hilum. Because there are more afferent vessels than efferent vessels, lymph backs up and moves slowly through the node. This gives the macrophages and lymphocytes time to work.

Structure	What it is
Capsule	Dense connective tissue covering; trabeculae extend inward from it
Afferent lymphatic vessels	Several vessels that carry lymph into the node through the capsule, on the convex surface
Subcapsular sinus	The space just under the capsule where lymph first enters
Cortex	Outer region; the outer cortex has lymphatic follicles of B cells with germinal centers, and the deep cortex is mostly T cells
Medulla	Inner region of medullary cords and medullary sinuses lined with macrophages
Hilum	The indented side of the node, where blood vessels enter and leave and efferent vessels leave
Efferent lymphatic vessels	Fewer vessels that carry filtered lymph out of the node at the hilum

The path of lymph through a node

1. Afferent lymphatic vessels: lymph enters through the capsule on the convex surface.
2. Subcapsular sinus: lymph spreads out in the space just under the capsule.
3. Cortical sinuses: lymph seeps through the cortex, past the follicles.
4. Medullary sinuses: macrophages lining these channels engulf bacteria, debris, and cancer cells.
5. Efferent lymphatic vessels: filtered lymph leaves at the hilum and moves on to the next node or a trunk.

What keeps lymph moving one way at each step

Structure	Wall or covering	What keeps lymph moving one way	Where lymph goes next
Lymphatic capillary	One layer of overlapping endothelial cells, held open by anchoring filaments	The overlapping cells act as flap valves	A collecting vessel
Collecting vessel	Built like a vein, with three thin layers	Many valves, more than veins have	A lymph node
Lymph node	A capsule of dense connective tissue	Lymph comes in only through the afferent vessels and leaves only through the efferent vessels	Out through the efferent vessels at the hilum, on toward a trunk

- Memory tip: afferent vessels arrive, efferent vessels exit. The same two words mean "in" and "out" in other systems too, such as the arterioles of the kidney and the neurons of the nervous system.
- Afferent vessels enter all around the capsule, but efferent vessels leave only at the hilum. In a drawing, more should come in than go out.

Competency 107. Primary and secondary organs

In your course notes: Module 4 packet, Chapter 2, competency 107 (page 22).

Dr. Rennie's video for this part: Lymphatic system concept video.

You should be able to: Sort red bone marrow, thymus, lymph nodes, spleen, tonsils, and MALT into primary and secondary lymphoid organs with location and role.

The main cells of the lymphatic system are lymphocytes, a type of white blood cell. The two kinds you need here are B cells and T cells. All lymphocytes are made in the red bone marrow. Before they can fight anything, they have to finish maturing. B cells mature in the red bone marrow. T cell precursors leave the marrow and travel in the blood to the thymus, where they mature.

This gives a simple way to sort the lymphoid (lymphatic) organs. Primary lymphoid organs are where lymphocytes are made and mature. There are only two: the red bone marrow and the thymus. Secondary lymphoid organs are where mature lymphocytes gather and meet pathogens. These are the lymph nodes, the spleen, the tonsils, and MALT. Mature lymphocytes leave the marrow and thymus and travel out to the secondary organs. Each secondary organ sits where it can catch pathogens from one source: the lymph nodes check lymph, the spleen checks blood, and the tonsils and MALT guard the linings of open tracts.

Organ	Group	Location	Role
Red bone marrow	Primary	Spongy bone of the axial skeleton (such as the sternum and vertebrae), the limb girdles (such as the hip bones), and the proximal ends of the femur and humerus	Where all lymphocytes form, and where B cells mature
Thymus	Primary	Mediastinum, above the heart and behind the sternum	Where T cells mature
Lymph nodes	Secondary	Along the collecting vessels, clustered in the cervical, axillary, and inguinal regions (neck, armpits, and groin)	Filter the lymph
Spleen	Secondary	Upper left abdomen, to the left of and behind the stomach, under the lower left ribs	Filters the blood
Tonsils	Secondary	A ring of lymphatic tissue in the walls of the pharynx and at the base of the tongue, around the entrance to the throat	Guard against inhaled and swallowed pathogens
MALT	Secondary	Mucous membranes of the digestive, respiratory, urinary, and reproductive tracts	Guards the mucosal surfaces

- Memory tip: only two organs are primary, the red bone marrow and the thymus, because those are the only places lymphocytes are made and matured. B cells mature in the bone marrow and T cells mature in the thymus. Everything else on the list is secondary.
- The spleen is the largest lymphatic organ, but it is secondary. Size does not make an organ primary; making and maturing lymphocytes does.

The thymus

The thymus sits in the mediastinum, above the heart and just behind the sternum. It is large in infants and children. After puberty it shrinks, and in older adults it is mostly fat. Its outer cortex is packed with immature T cells. Its inner medulla has fewer, more mature T cells and small round structures called thymic corpuscles. The thymus does not filter lymph. Unlike a lymph node, it has no afferent lymphatic vessels and no follicles.

The spleen

The spleen is the largest lymphatic organ. It sits in the upper left abdomen against the diaphragm, to the left of and behind the stomach, and it is protected by the lower left ribs. The splenic artery and splenic vein pass through its hilum. The spleen filters the blood. It removes worn-out red blood cells and platelets and responds to pathogens carried in the blood. Its tissue comes in two kinds.

Structure	What it is
Spleen	The largest lymphatic organ, in the upper left abdomen under the lower left ribs
White pulp	Small islands of lymphocytes clustered around central arteries; carries out the spleen's immune functions
Red pulp	Most of the organ; venous sinuses and splenic cords full of red blood cells and macrophages that filter the blood and remove worn-out erythrocytes

The tonsils and MALT

MALT stands for mucosa-associated lymphatic tissue. It is lymphatic tissue in the mucous membranes that line the open tracts of the body, where pathogens can get in. The largest and best organized pieces of MALT are the tonsils, the Peyer patches, and the appendix.

The tonsils form a ring of lymphatic tissue around the entrance to the pharynx, where they meet inhaled and swallowed material. Their surface has deep pits called crypts that trap bacteria and debris.

Structure	Location	Notes
Pharyngeal tonsil	Posterior wall of the nasopharynx	Single; also called the adenoid
Palatine tonsils	Lateral walls of the oropharynx, between the palatoglossal and palatopharyngeal arches	Paired and the largest; the ones usually infected and removed
Lingual tonsils	Base (posterior third) of the tongue, facing the oropharynx	Guard swallowed material
Tubal tonsils	Around the openings of the auditory tubes in the nasopharynx	Paired
Peyer patches	Wall of the ileum, the last part of the small intestine	Clusters of lymphatic follicles that watch the intestinal contents
Appendix	Off the cecum of the large intestine	Wall packed with lymphatic follicles

Competency 108. Lymphatic disorders

In your course notes: Module 4 packet, Chapter 2, competency 108 (page 24).

You should be able to: Name edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis and identify the structure each one affects.

Lymphatic disorders fall into two groups. In the first group, drainage fails and fluid collects in the tissues, which shows up as swelling. In the second group, the lymphatic tissue itself changes. A node, the spleen, or the tonsils become enlarged, inflamed, or cancerous.

Edema is swelling caused by excess fluid in the interstitial space. It happens when fluid leaks out of the blood capillaries faster than the lymphatic vessels can carry it away. Lymphedema is a specific kind of edema. It is long-term swelling caused by blocked, damaged, or missing lymphatic vessels. It can follow surgery that removes lymph nodes, because the vessels that passed through those nodes no longer drain the region. The difference between the two is where the problem sits: edema is a problem in the tissue space, while lymphedema is a problem in the lymphatic vessels themselves.

Lymphadenopathy means enlarged lymph nodes. Nodes often swell when they are fighting an infection nearby, and they can also enlarge when cancer cells lodge in them. Lymphoma is a cancer of the lymphocytes in lymphatic tissue. It usually starts in lymph nodes and comes in two main groups, Hodgkin lymphoma and non-Hodgkin lymphoma. Lymphadenopathy is any enlarged node, for any reason; lymphoma is a cancer of the lymphocytes. Splenomegaly is an enlarged spleen. It can happen with an infection such as mononucleosis or with a blood disorder such as leukemia. Tonsillitis is inflammation of the tonsils, usually the palatine tonsils, and it is common in childhood.

Disorder	Structure affected	What changes
Edema	The interstitial space	Excess interstitial fluid builds up and the tissue swells, because fluid leaks out faster than the lymphatics can drain it
Lymphedema	The lymphatic vessels	Blocked, damaged, or missing vessels cause long-term swelling of the region they drain, sometimes after lymph node removal
Lymphadenopathy	The lymph nodes	The nodes enlarge, often while fighting an infection, and sometimes because cancer cells have lodged in them
Lymphoma	Lymphatic tissue, usually the lymph nodes	Lymphocytes become cancerous and the tissue enlarges; Hodgkin or non-Hodgkin lymphoma
Splenomegaly	The spleen	The spleen enlarges, for example in mononucleosis or leukemia, and becomes easier to rupture
Tonsillitis	The tonsils, usually the palatine tonsils	The tonsils become infected, red, and swollen

What this means for patients

- Swollen glands are lymphadenopathy. Lymph nodes enlarge as they trap and fight pathogens, which is why a clinician feels for tender nodes in the neck, armpit, and groin during an exam.
- Because lymph passes through nodes in order, the lymphatic system also maps how some cancers spread. The first node a region drains into is the sentinel node, and checking it tells a surgeon whether a cancer has begun to travel. Removing many nodes can block drainage and leave lasting lymphedema in the limb.
- The spleen sits just under the lower left ribs. Its soft tissue tears easily, and it holds a large volume of blood, so a blow to that area can rupture the spleen and cause heavy internal bleeding. An enlarged spleen, as in mononucleosis, ruptures even more easily, which is why contact sports are dangerous during mononucleosis.

Worked questions

Eight questions in the style of the lecture exam, two for each competency. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 105

The thoracic duct empties lymph into the blood at which point?

A. The junction of the right subclavian and right internal jugular veins

B. The cisterna chyli

C. The superior vena cava

D. The junction of the left subclavian and left internal jugular veins

Answer: D. Both ducts empty where a subclavian vein meets an internal jugular vein, each on its own side. The thoracic duct ends on the left, so it empties into the junction of the left subclavian and left internal jugular veins.

Why the others are wrong

- **A** That is where the right lymphatic duct empties. The thoracic duct ends on the left side of the base of the neck.
- **B** The cisterna chyli is where the thoracic duct begins, a sac in the abdomen in front of the upper lumbar vertebrae. Lymph enters the duct there; it does not enter the blood there.
- **C** The ducts end at the vein junctions in the base of the neck. Blood from those junctions flows on into the superior vena cava, but the ducts do not open into it directly.

QUESTION 2 · COMPETENCY 105

Lymph from the right leg returns to the blood through which vessel?

A. The right lymphatic duct

B. The thoracic duct

C. The right subclavian trunk

D. The right jugular trunk

Answer: B. The right lymphatic duct drains only the right upper quarter of the body: the right arm and the right side of the head, neck, and thorax. Every other region, including both legs, drains through the lumbar trunks into the cisterna chyli and up the thoracic duct.

Why the others are wrong

- **A** The right lymphatic duct is short and small and takes only the right jugular, right subclavian, and right bronchomediastinal trunks. The right leg is not in its territory.
- **C** The subclavian trunks drain the upper limbs. The right leg drains into the right lumbar trunk.
- **D** The jugular trunks drain the head and neck, not a limb.

QUESTION 3 · COMPETENCY 106

Which vessels carry lymph out of a lymph node, and where do they leave?

A. Efferent vessels, leaving only at the hilum

B. Afferent vessels, leaving only at the hilum

C. Efferent vessels, leaving through the capsule all over the convex surface

D. Afferent vessels, leaving through the capsule all over the convex surface

Answer: A. Efferent vessels exit. Filtered lymph leaves a node through a few efferent vessels at the hilum, the indented side where the blood vessels also enter and leave. Because several afferent vessels bring lymph in and fewer efferent vessels take it out, lymph slows down inside the node.

Why the others are wrong

- **B** Afferent vessels arrive. They carry lymph into the node, and they do not use the hilum.
- **C** Lymph enters through the capsule on the convex surface; it does not leave there. Efferent vessels are found only at the hilum.
- **D** Afferent vessels do enter through the capsule on the convex surface, but they carry lymph in, not out.

QUESTION 4 · COMPETENCY 106

What holds a lymphatic capillary open when the surrounding tissue swells with fluid?

A. The pressure of the lymph inside it

B. A thick layer of smooth muscle in its wall

C. Anchoring filaments tied to the surrounding tissue

D. Valves in its lumen

Answer: C. Anchoring filaments are fine collagen strands that tie the endothelial cells of the lymphatic capillary to the tissue around it. When the tissue swells, the filaments pull on the cells and hold the capillary open, so fluid can flow in instead of the tube collapsing.

Why the others are wrong

- **A** Internal pressure holds a blood capillary open. In a lymphatic capillary, rising pressure inside presses the overlapping flaps shut; it is not what keeps the tube open.
- **B** A lymphatic capillary wall is a single layer of endothelial cells with no muscle. Smooth muscle appears in the walls of the larger lymphatic vessels.
- **D** Lymphatic capillaries have no internal valves. Their one-way flow comes from the overlapping endothelial cells acting as flap valves. Internal valves are a feature of the collecting vessels.

QUESTION 5 · COMPETENCY 107

Which two organs are the primary lymphoid organs?

A. Spleen and thymus

B. Red bone marrow and thymus

C. Red bone marrow and spleen

D. Lymph nodes and thymus

Answer: B. Primary lymphoid organs are where lymphocytes are made and mature. All lymphocytes form in the red bone marrow, B cells mature there, and T cells mature in the thymus. Those are the only two places that happens.

Why the others are wrong

- **A** The thymus is primary, but the spleen is secondary. It filters blood and is where mature lymphocytes meet pathogens; it does not make or mature them. Being the largest lymphatic organ does not make it primary.
- **C** Red bone marrow is primary, but the spleen is secondary for the same reason.
- **D** The thymus is primary, but lymph nodes are secondary. They filter lymph and hold mature lymphocytes.

QUESTION 6 · COMPETENCY 107

Which lymphoid organ filters the blood, removing worn-out red blood cells and platelets?

A. Thymus

B. Lymph node

C. Palatine tonsil

D. Spleen

Answer: D. The spleen filters blood. Its red pulp is full of venous sinuses, splenic cords, and macrophages that remove worn-out erythrocytes and platelets, and its white pulp responds to pathogens carried in the blood. It sits in the upper left abdomen under the lower left ribs.

Why the others are wrong

- **A** The thymus is where T cells mature. It does not filter blood or lymph, and it has no afferent lymphatic vessels and no follicles.
- **B** A lymph node filters lymph, not blood. Each secondary organ checks one source, and nodes check the lymph that passes through them.
- **C** The palatine tonsils are part of the ring of lymphatic tissue around the entrance to the pharynx. They guard against inhaled and swallowed pathogens; they do not filter blood.

QUESTION 7 · COMPETENCY 108

A patient has long-term swelling of one arm after surgery that removed the axillary lymph nodes. Which disorder is this, and which structure is affected?

A. Edema; the interstitial space

B. Lymphadenopathy; the lymph nodes

C. Lymphedema; the lymphatic vessels

D. Lymphoma; lymphatic tissue

Answer: C. Lymphedema is long-term swelling caused by blocked, damaged, or missing lymphatic vessels. Removing the axillary nodes cuts the vessels that passed through them, so the arm no longer drains and the swelling lasts.

Why the others are wrong

- **A** Edema is swelling in the interstitial space because fluid leaks out faster than healthy lymphatics can drain it. Here the problem is the lymphatic vessels themselves, which makes it lymphedema, a specific kind of edema.
- **B** Lymphadenopathy is enlarged lymph nodes. In this patient the nodes have been removed, not enlarged.
- **D** Lymphoma is a cancer of the lymphocytes in lymphatic tissue. Nothing here is cancerous; the swelling is a drainage problem.

QUESTION 8 · COMPETENCY 108

A student with mononucleosis is told to avoid contact sports for several weeks. Which disorder makes this necessary, and which organ is at risk?

A. Splenomegaly; the spleen could rupture

B. Tonsillitis; the tonsils could rupture

C. Lymphadenopathy; the neck nodes could rupture

D. Lymphoma; the lymph nodes could rupture

Answer: A. Mononucleosis can enlarge the spleen, which is splenomegaly. The spleen sits just under the lower left ribs, its soft tissue tears easily, and it holds a large volume of blood, so a blow to that area can rupture it and cause heavy internal bleeding. An enlarged spleen ruptures even more easily.

Why the others are wrong

- **B** Tonsillitis is inflammation of the tonsils, usually the palatine tonsils. It can be part of an illness, but it is not the reason to avoid contact sports.
- **C** Lymphadenopathy is enlarged lymph nodes, which do occur in infections, but the structure at risk from a blow is the spleen, not the nodes.
- **D** Lymphoma is a cancer of the lymphocytes. Mononucleosis is an infection, not a cancer.